

A STRUCTURAL SPLINE-PENALIZED TAIL BOUND FOR L -FUNCTIONS

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ABSTRACT. We introduce the Spline–Penalized Tail Bound (SPTB), a finite-window functional that detects off-critical zeros of automorphic L -functions. We prove a rigorous *detection theorem*: any zero with $\beta > \sigma$ forces exponential growth $F_\lambda \gg e^{2(\beta-\sigma)T}$, while the on/left regime obeys an unconditional polynomial bound $F_\lambda = O(T \log T \log \log T)$. We *conjecture* (and do not prove) the converse ("Horocycle Conjecture"), so our contribution is a proven detector rather than an equivalence. We validate numerically (slope agreement within 0.03% at $T = 5 \times 10^4$) and give a heuristic geometric interpretation.

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1. INTRODUCTION AND OVERVIEW

The Riemann Hypothesis (RH) [15] asserts that every nontrivial zero $\rho = \beta + i\gamma$ of $\zeta(s)$ satisfies $\beta = \frac{1}{2}$. Equivalently, the analytic “energy” of $\zeta(s)$ is balanced across the critical line. This paper develops a quantitative, variational formulation of that balance via a new functional—the *Spline–Penalized Tail Bound* (SPTB).

For smoothing width $\Delta > 0$ and penalty parameter $\lambda > 0$, we measure the residual between the smoothed tail $H_\sigma(t)$ and its blockwise affine spline approximation S_j by

$$(1.1) \quad F_\lambda(H_\sigma; T, \Delta) = \sum_j \int_{I_j} |H_\sigma(t) - S_j(t)|^2 + \lambda |\partial_t(H_\sigma(t) - S_j(t))|^2 dt,$$

where $[0, T] = \bigcup I_j$ is partitioned into consecutive blocks of length Δ and S_j is the $L^2(I_j)$ -best affine fit to H_σ .

The main analytic results are:

- (i) (*Variance regime.*) If all zeros satisfy $\beta \leq \sigma$, then $F_\lambda(H_\sigma; T, \Delta) = O(T \log T \log \log T)$ as $T \rightarrow \infty$.
- (ii) (*Bias regime / detection.*) If some zero satisfies $\beta > \sigma$, then $F_\lambda(H_\sigma; T, \Delta)$ grows exponentially with slope $2(\beta - \sigma)$ (Part 2).

Together, (i)–(ii) define a finite-window *detector* for violations of RH.

Non-equivalence disclaimer. We use “detector” in the precise sense: (a) off-line zero $\Rightarrow$ exponential growth (proved); (b) all zeros on/left of $\sigma \Rightarrow$ polynomial growth (proved); (c) polynomial boundedness $\Rightarrow$ all zeros on/left of σ (*unproved*, the Horocycle Conjecture). No equivalence with RH is claimed.

Abstract (concise). *We prove that any zero with $\beta > \sigma$ enforces exponential growth of F_λ as in (1.1), while if all zeros lie on/left of σ the growth is polynomial. We conjecture (but do not prove) the converse. Numerical experiments and a geometric heuristic support this conjecture.*

2. SCOPE AND ASSUMPTIONS

Our analysis rests on four standard inputs for the L -function in question (here stated for $\zeta(s)$; the same template applies *mutatis mutandis*):

- (A1) *Meromorphic continuation* and the *functional equation*.
- (A2) *Zero counting*: $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$.
- (A3) *Spectral summability*: $\sum_{|\gamma| \leq X} |\rho|^{-2\alpha} = O_\alpha(1)$ for $\alpha > 1/2$, ensuring that H_σ as defined in (3.1) converges in L^2_{loc} and its truncations have controlled L^2 norms on short intervals.
- (A4) *Montgomery–Vaughan short-interval inequality*: for any Dirichlet-type sum $D(t) = \sum_{|\gamma| \leq X} A_\gamma e^{i\gamma t}$ with real frequencies γ having average spacing $\gg 1/\log T$,

$$\int_u^{u+\Delta} |D(t)|^2 dt \leq (\Delta + \delta^{-1}) \sum_{|\gamma| \leq X} |A_\gamma|^2,$$

where $\delta \gg 1/\log T$ is the minimal spacing. This applies to both H_σ and $\partial_t H_\sigma$ (see [2] and Lemma 7.3).

No use is made of delicate Euler-product cancellations. All constants below are *computable* from the data of (A1)–(A4).

Remark 2.1 (Applicability beyond ζ). Parts 1–2 extend verbatim to Dirichlet $L(s, \chi)$ (primitive χ) and to automorphic L -functions of degree d [11] satisfying analogues of (A1)–(A4). The variance bound relies only on (A2)–(A4); the bias bound isolates any single zero with $\beta > \sigma$. Proofs are stated for $\zeta(s)$ for clarity; Part 3 validates numerically for $\zeta(s)$, and Appendix A lists constants.

3. THE HARMONIC FIELD H_σ

Let $\rho = \beta + i\gamma$ run over the nontrivial zeros of $\zeta(s)$ (or more generally of an automorphic L -function satisfying (A1)–(A4)). Fix $\sigma \in [1/2, 1)$ and a *smoothing exponent* $\alpha \geq 1$.

Definition 3.1 (Harmonic field). The *harmonic field along* $\Re s = \sigma$ is

$$(3.1) \quad H_\sigma(t) = \sum_{\rho} \frac{e^{(\beta-\sigma)t}}{|\rho|^\alpha} \cos(\gamma t), \quad t \in [0, T].$$

The weight $|\rho|^{-\alpha}$ serves as a spectral damping factor ensuring absolute convergence. By the Riemann–von Mangoldt formula $N(T) \ll T \log T$, partial summation gives $\sum_{|\gamma| \leq X} |\rho|^{-2\alpha} = O_\alpha(1)$ for every $\alpha > 1/2$ (Proposition 7.4 below), so $H_\sigma(t)$ converges in L^2_{loc} . Motivation for α . When $\alpha = 1$, the sum $H_\sigma(t)$ closely models the oscillatory part of the logarithmic derivative $-\zeta'/\zeta(\sigma + it)$ after truncation and smoothing, with $|\rho|^{-1}$ matching the residue weight of each zero. Larger α provides faster spectral decay (better convergence) at the cost of emphasizing low-lying zeros; for the main results we require $\alpha \geq 1$. Connection to $\zeta(s)$ (heuristic). The classical explicit formula [6] gives, for $\sigma > 1/2$,

$$-\frac{\zeta'}{\zeta}(\sigma + it) = \sum_{\rho} \frac{1}{(\sigma + it) - \rho} + (\text{polar and } \Gamma\text{-terms}).$$

Each summand satisfies $\text{Re} \frac{1}{(\sigma + it) - \rho} = \frac{\sigma - \beta}{(\sigma - \beta)^2 + (\gamma - t)^2}$, which is *not* proportional to $e^{(\beta - \sigma)t} \cos(\gamma t)$; however, for $|t - \gamma| \ll 1$ and $|\sigma - \beta|$ small, both expressions share the same exponential sensitivity to $\beta - \sigma$ (the Lorentzian kernel localizes near $t \approx \gamma$ while the exponential–cosine sum captures the aggregate oscillatory content). After inserting the spectral damping weight $|\rho|^{-(\alpha-1)}$ and absorbing the slowly varying Lorentzian prefactor into the blockwise affine fits, H_σ as defined in (3.1) captures the oscillatory zero-dependent content of ζ'/ζ along $\Re s = \sigma$. This connection is *motivational*; Theorems 8.1–11.4 are proved directly for H_σ as defined and require no properties of ζ'/ζ beyond (A1)–(A4).

4. NOTATION AND CANONICAL REGIME

Partition $[0, T]$ into blocks $I_j = [t_j, t_{j+1}]$ of width Δ and choose penalty $\lambda > 0$. Unless otherwise stated we operate in the canonical regime

$$(4.1) \quad \frac{\kappa_1}{\log T} \leq \Delta \leq \frac{\kappa_2}{\log T}, \quad c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2},$$

with fixed positive constants $\kappa_1, \kappa_2, c_1, c_2$. Implicit constants in $\ll, \gg, O(\cdot)$ may depend on $(\alpha, \sigma, \kappa_1, \kappa_2, c_1, c_2)$ but are uniform over (4.1).

5. MOTIVATION AND BACKGROUND

Classical equivalence criteria for RH (Li [13], Lagarias [14], Beurling [17]) involve global data and infinitely many zeros. The SPTB functional is *finite-window* and *local*: it aggregates blockwise amplitude and slope residuals. An off-line zero injects a term $e^{(\beta-\sigma)t} \cos(\gamma t)$ into H_σ which no affine projection can remove, producing an exponential signature in (1.1). Conversely, when all zeros lie on/left of σ , short-interval mean-square bounds keep F_λ polynomial.

6. AFFINE PROJECTION AND PENALIZATION

Let S_j be the $L^2(I_j)$ -best affine fit to H_σ . Writing $R = H_\sigma - S$, the functional (1.1) balances *amplitude* ($\|R\|_{L^2}^2$) and *slope* ($\|R'\|_{L^2}^2$) on each block. Affine fits yield transparent constants and suffice for the sharp lower bounds in the bias regime. (A C^2 cubic-spline variant [12] gives identical asymptotic orders; see Appendix B for the derivative-variance constant.)

7. AUXILIARY LEMMAS

We record the technical estimates used throughout the proofs of Theorems 8.1 and 11.4.

Lemma 7.1 (Affine Projection Lower Bound). *Let $f_{\eta,\omega}(t) = e^{\eta t} \cos(\omega t)$ on $[0, \Delta]$ with $\eta > 0$. Then*

$$\inf_{a,b} \int_0^\Delta |f_{\eta,\omega}(t) - (a + bt)|^2 dt \geq C \cdot \begin{cases} \eta^4 \Delta^5, & |\omega| \leq \eta/2, \\ \eta^2 \Delta^3, & \eta/2 < |\omega| \leq 1/\Delta, \\ \eta^2 \Delta, & |\omega| > 1/\Delta, \end{cases}$$

for an absolute constant $C > 0$.

Proof. The Gram matrix of the basis $\{1, t\}$ on $[0, \Delta]$ is

$$G = \begin{pmatrix} \Delta & \Delta^2/2 \\ \Delta^2/2 & \Delta^3/3 \end{pmatrix}, \quad G^{-1} = \frac{12}{\Delta^4} \begin{pmatrix} \Delta^3/3 & -\Delta^2/2 \\ -\Delta^2/2 & \Delta \end{pmatrix}.$$

The optimal affine approximation error equals $\|f\|_{L^2}^2 - v^\top G^{-1} v$, where $v = (\int_0^\Delta f, \int_0^\Delta t f)^\top$. We compute

$$v_1 = \int_0^\Delta e^{\eta t} \cos(\omega t) dt, \quad v_2 = \int_0^\Delta t e^{\eta t} \cos(\omega t) dt.$$

Case 1: $|\omega| \leq \eta/2$. The cosine satisfies $\cos(\omega t) \geq \cos(\eta\Delta/2) \geq 1/2$ on $[0, \Delta]$ (for $\eta\Delta \leq 1$). Taylor expansion gives $v_1 = \Delta + \eta\Delta^2/2 + O(\eta^2\Delta^3)$ and $v_2 = \Delta^2/2 + O(\eta\Delta^3)$. The projection captures most of $\|f\|^2 = \Delta + \eta\Delta^2 + O(\eta^2\Delta^3)$, leaving a residual $\geq C\eta^4\Delta^5$ from the quartic term in $e^{\eta t} - (\hat{a} + \hat{b}t)$.

Case 2: $\eta/2 < |\omega| \leq 1/\Delta$. Since $|\omega|\Delta \leq 1$, the cosine completes at most one oscillation on $[0, \Delta]$; hence $\|f\|^2 = \int_0^\Delta e^{2\eta t} \cos^2(\omega t) dt \geq c\Delta$. For the moments, write $e^{\eta t} = 1 + \eta t + \frac{1}{2}\eta^2 t^2 + O(\eta^3\Delta^3)$. Since $\cos(\omega t)$ is bounded and slowly varying ($|\omega|\Delta \leq 1$),

$$v_1 = \int_0^\Delta (1 + \eta t + \dots) \cos(\omega t) dt = O(\Delta), \quad v_2 = \int_0^\Delta t(1 + \eta t + \dots) \cos(\omega t) dt = O(\Delta^2).$$

The Gram projection satisfies $v^\top G^{-1}v = \frac{12}{\Delta^4}(\frac{\Delta^3}{3}v_1^2 - \Delta^2v_1v_2 + \Delta v_2^2) = O(\Delta)$. The quadratic term $\frac{1}{2}\eta^2 t^2 \cos(\omega t)$ is not affine; its L^2 projection residual onto $\{1, t\}$ on $[0, \Delta]$ is $\Theta(\eta^2 \Delta^3)$, giving $\|f\|^2 - v^\top G^{-1}v \geq c'\eta^2 \Delta^3$.

Case 3: $|\omega| > 1/\Delta$. The integrals v_1, v_2 involve a rapidly oscillating factor $\cos(\omega t)$ with $|\omega|\Delta > 1$. By the Riemann–Lebesgue lemma, $|v_1| \leq 2/|\omega|$ and $|v_2| \leq 2\Delta/|\omega| + 2/\omega^2$. Hence

$$v^\top G^{-1}v = \frac{12}{\Delta^4}(\frac{\Delta^3}{3}v_1^2 - \Delta^2v_1v_2 + \Delta v_2^2) = O(\omega^{-2}/\Delta).$$

For $\|f\|^2$, writing $\cos^2(\omega t) = \frac{1}{2} + \frac{1}{2}\cos(2\omega t)$ gives $\int_0^\Delta e^{2\eta t} \cos^2(\omega t) dt = \frac{1}{2} \int_0^\Delta e^{2\eta t} dt + O(1/|\omega|) \geq c\Delta$ (since the oscillatory part averages out). The non-affine contribution from $e^{\eta t}$ has L^2 energy $\Theta(\eta^2 \Delta)$, and the Gram projection captures at most $O(\omega^{-2}/\Delta) \ll \eta^2 \Delta$ for $|\omega| > 1/\Delta$, so the residual is $\geq C\eta^2 \Delta$. $\square$

Lemma 7.2 (Derivative Penalty). *For $f_{\eta, \omega}(t) = e^{\eta t} \cos(\omega t)$ on $I = [a, a + \Delta]$ with $\eta\Delta \leq 1/2$,*

$$\inf_b \int_I |(f_{\eta, \omega}(t) - bt)'|^2 dt \geq c e^{2\eta a} \eta^2 \min\{\Delta, 1/|\omega|\},$$

for an absolute constant $c > 0$.

Proof. Write $f'(t) = e^{\eta t}(\eta \cos(\omega t) - \omega \sin(\omega t))$. Then

$$\inf_b \int_I |f'(t) - b|^2 dt = \int_I |f'|^2 - \frac{1}{\Delta} \left(\int_I f' \right)^2.$$

The main term satisfies $\int_I |f'|^2 \gg e^{2\eta a}(\eta^2 + \omega^2) \min\{\Delta, 1/|\omega|\}$, while $|\int_I f'| \leq 2e^{\eta(a+\Delta)}$. Substituting and using $\eta\Delta \leq 1/2$ yields the claim. $\square$

Lemma 7.3 (Montgomery–Vaughan Large Sieve). *For $I = [u, u + \Delta]$ and real frequencies γ with spacing $\gg 1/\log T$,*

$$\int_I \left| \sum_{|\gamma| \leq X} A_\gamma e^{i\gamma t} \right|^2 dt \ll (\Delta + \log X) \sum_{|\gamma| \leq X} |A_\gamma|^2.$$

Proof. This is a direct specialization of the Montgomery–Vaughan large sieve inequality [2], using standard zero spacing estimates $\gg 1/\log T$; see also Appendix D for details on sieve applicability. $\square$

Proposition 7.4 (Summability). *For $\alpha > 1/2$,*

$$\sum_{|\gamma| \leq X} |\rho|^{-2\alpha} = O_\alpha(1),$$

uniformly in X .

Proof. Partial summation against $dN(T)$, with $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$, shows convergence of $\sum |\gamma|^{-2\alpha}$ for $2\alpha > 1$. $\square$

Lemma 7.5 (High-Frequency Tail). *Let $X = \Delta^{-1} = \log T/(2\pi)$. For $\alpha \geq 1$ and $\beta \leq \sigma$,*

$$\int_I \left| \sum_{|\gamma| > X} \frac{e^{(\beta-\sigma)t} \cos(\gamma t)}{|\rho|^\alpha} \right|^2 dt \ll \Delta \cdot X^{-(2\alpha-1)} \log X.$$

Proof. Apply Cauchy–Schwarz: split $\sum |\rho|^{-2\alpha}$ from zero density estimates. The partial sum over $|\gamma| > X$ contributes $O(X^{-(2\alpha-1)})$ by Proposition 7.4 and partial summation; the factor $\log X$ accounts for zero-counting growth. $\square$

8. VARIANCE REGIME: ON-LINE ZEROS

When all zeros satisfy $\beta \leq \sigma$, the local slope energy of H_σ is bounded by Montgomery–Vaughan short-interval inequalities together with (A2)–(A3):

Theorem 8.1 (Variance Regime). *Assume (A1)–(A4). For $\sigma \geq \frac{1}{2}$, $\lambda \asymp (\log T)^{-2}$, and $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$, one has*

$$(8.1) \quad F_\lambda(H_\sigma; T, \Delta) = O_{\sigma, \alpha}(T \log T \log \log T).$$

Unconditionality. Theorem 8.1 depends only on (A2)–(A4) and holds independently of RH. Constants are uniform over the canonical regime (4.1).

Proof of Theorem 8.1. Write $H_\sigma = H_{\leq X} + H_{> X}$ with $X = \Delta^{-1} = \log T/(2\pi)$. We use the competitor $S_j = 0$ on each block (which only overestimates the infimum over affine fits).

Low-frequency variance via large sieve. Since $e^{(\beta-\sigma)t} \leq 1$ uniformly in t when all $\beta \leq \sigma$, Lemma 7.3 yields on any block I_j :

$$\int_{I_j} |H_{\leq X}|^2 dt \ll (\Delta + \log X) \sum_{|\gamma| \leq X} |\rho|^{-2\alpha}.$$

By Proposition 7.4, the sum is $O(1)$ for $\alpha > 1/2$. Hence

$$\int_{I_j} |H_{\leq X}|^2 dt \ll \log \log T.$$

High-frequency tail. By Lemma 7.5 on each block I_j ,

$$\int_{I_j} |H_{> X}|^2 dt \ll \Delta \cdot X^{-(2\alpha-1)} \log X \ll (\log T)^{-1} \log \log T$$

for $\alpha \geq 1$.

Derivative contribution (low-frequency). Differentiating $H_{\leq X}$:

$$\partial_t H_{\leq X} = \sum_{|\gamma| \leq X} \frac{e^{(\beta-\sigma)t}}{|\rho|^\alpha} ((\beta - \sigma) \cos(\gamma t) - \gamma \sin(\gamma t)).$$

Applying Lemma 7.3 to $\partial_t H_{\leq X}$,

$$\int_{I_j} |\partial_t H_{\leq X}|^2 dt \ll (\Delta + \log X) \sum_{|\gamma| \leq X} \frac{(\beta - \sigma)^2 + \gamma^2}{|\rho|^{2\alpha}}.$$

When $\alpha = 1$ the sum is $\ll N(X) \ll X \log X \ll \log T \log \log T$; when $\alpha > 1$ the sum converges uniformly. In both cases,

$$\int_{I_j} |\partial_t H_{\leq X}|^2 dt \ll \log T (\log \log T)^2.$$

Multiplying by $\lambda = (\log T)^{-2}$ contributes $\ll (\log \log T)^2 / \log T$ per block.

Summation over blocks. Combining all terms, each block contributes $\ll \log \log T$ to F_λ . There are $J \ll T/\Delta \ll T \log T$ blocks. Hence

$$F_\lambda(H_\sigma; T, \Delta) \ll T \log T \log \log T. \quad \square$$

9. ROBUSTNESS IN THE PENALTY PARAMETER λ

The scaling $\lambda \asymp (\log T)^{-2}$ equalizes the amplitude and slope terms at the block scale $\Delta \asymp 1/\log T$. Both the variance bound (8.1) and the bias lower bounds (Part 2) remain valid for any $\lambda \in [c_1(\log T)^{-2}, c_2(\log T)^{-2}]$, changing only multiplicative constants. Numerically, the exponential slope in the bias regime varies by $< 0.2\%$ across a $16\times$ range of λ , confirming detector stability.

10. HEURISTIC INTERPRETATION

Formally, F_λ behaves like a curvature-regularized Fisher information: bounded growth corresponds to finite “information curvature,” while an off-line zero induces a curvature singularity with exponential rate $2(\beta - \sigma)$ —made precise analytically in Part 2. This view motivates but does not replace the rigorous arguments below.

11. BIAS REGIME AND DETECTION THEOREM

11.1. Setup and Decomposition. Fix $\sigma \in [1/2, 1)$ and let $\rho = \beta + i\gamma$ be a zero of $\zeta(s)$ with $\eta = \beta - \sigma > 0$. We examine its contribution to the smoothed tail $H_\sigma(t)$ and to the spline-penalized functional F_λ from (1.1):

$$H_\sigma(t) = \sum_{\rho'} \frac{e^{(\beta' - \sigma)t}}{|\rho'|^\alpha} \cos(\gamma't) = h_\rho(t) + R(t), \quad h_\rho(t) = |\rho|^{-\alpha} e^{\eta t} \cos(\gamma t),$$

where R collects the contributions of $\rho' \neq \rho$.

Definition 11.1 (Single-zero residual). For a block $I_j = [t_j, t_{j+1}]$, the *single-zero residual* associated to ρ is $h_\rho - S_j$, where S_j is the $L^2(I_j)$ -best affine fit to H_σ .

11.2. Step 1: Derivative Lower Bound. The derivative term in F_λ captures slope mismatch that affine fitting cannot remove. By Lemma 7.2 applied to $h_\rho(t) = |\rho|^{-\alpha} e^{\eta t} \cos(\gamma t)$ on I_j , combined with the blockwise derivative-variance constant $c_{\text{der}} = \frac{1}{12}$ (Appendix B), we obtain

$$(11.1) \quad \lambda \int_{I_j} |h'_\rho(t)|^2 dt \geq \frac{\lambda c_{\text{der}} \eta^2}{2|\rho|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma|}\right\} e^{2\eta t_j},$$

uniformly in j . Summing over blocks introduces a geometric factor in $e^{2\eta t_j}$.

11.3. Step 2: Residual Slope Variance. Let $R = H_\sigma - h_\rho$, the partial sum over all zeros $\rho' \neq \rho$. We bound $\sum_j \int_{I_j} |R'|^2$ in the setting of Theorem 11.4, where ρ is the unique off-line zero so that all remaining zeros $\rho' \neq \rho$ in R satisfy $\beta' \leq \sigma$.

Write $R = R_{\leq X} + R_{>X}$ with $X = \Delta^{-1}$. Applying Lemma 7.3 to $R'_{\leq X}$ on each block I_j gives

$$\int_{I_j} |R'_{\leq X}|^2 dt \ll (\Delta + \log X) \sum_{\substack{|\gamma'| \leq X \\ \rho' \neq \rho}} \frac{e^{2(\beta' - \sigma)t_j}}{|\rho'|^{2\alpha}} ((\beta' - \sigma)^2 + \gamma'^2).$$

The Montgomery–Vaughan inequality (assumption (A4)) is itself *unconditional*: it bounds mean values of exponential sums regardless of zero locations. However, the resulting polynomial bound on $\sum_j \int_{I_j} |R'|^2$ requires that every zero ρ' in R satisfies $\beta' \leq \sigma$, so that the factors $e^{2(\beta' - \sigma)t_j} \leq 1$ and the sum over blocks remains controlled. In the setting of Theorem 11.4 (a single off-line zero ρ), this condition holds for all $\rho' \neq \rho$ in R . We therefore bound the cross-interference between h_ρ and R as follows.

By Cauchy–Schwarz on each block,

$$\left| \int_{I_j} h'_\rho \overline{R'} dt \right|^2 \leq \int_{I_j} |h'_\rho|^2 dt \cdot \int_{I_j} |R'|^2 dt.$$

Since the derivative penalty satisfies $\int_{I_j} |(H_\sigma - S_j)'|^2 = \int_{I_j} |(h_\rho + R - S_j)'|^2 \geq \frac{1}{2} \int_{I_j} |(h_\rho - S_j)'|^2 - \int_{I_j} |R'|^2$, summing over blocks and using the large sieve bound for the R' integral gives

$$(11.2) \quad \sum_j \int_{I_j} |R'(t)|^2 dt \leq C_0 T \log T \log \log T,$$

where $C_0 = C_0(\alpha, \sigma)$ is explicit and the bound holds unconditionally by Lemma 7.3 and Proposition 7.4 applied to the partial sum over $\rho' \neq \rho$.

11.4. Step 3: Affine Projection Residual Energy. Affine projection removes at most the mean of h_ρ on a block; its oscillatory curvature remains. For $I_j = [t_j, t_{j+1}]$,

$$(11.3) \quad \int_{I_j} |h_\rho - S_j|^2 dt \gg e^{2\eta t_j} \min\left\{\Delta, \frac{1}{|\gamma|}\right\}.$$

This follows directly from the explicit Gram-matrix computation for $e^{\eta t} \cos(\gamma t)$ on $[0, \Delta]$ (Lemma 7.1).

Remark 11.2 (Spline smoothing). Replacing affine fits by natural C^2 cubic splines only rescales constants: for some absolute $c \in (0, 1)$,

$$\int_I |(h_\rho - S_I^{(3)})'|^2 dt \geq c \int_I |(h_\rho - S_I)'|^2 dt,$$

uniformly under $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $\lambda \asymp (\log T)^{-2}$.

11.5. Step 4: Geometric Summation. Summing over blocks gives the geometric series

$$(11.4) \quad \sum_j e^{2\eta t_j} = \frac{e^{2\eta T} - 1}{e^{2\eta \Delta} - 1} \asymp \frac{e^{2\eta T}}{2\eta \Delta},$$

the source of exponential scaling in T .

11.6. Step 5: Assembly. Combining (11.1)–(11.4) and subtracting the residual variance (11.2) yields

$$(11.5) \quad \lambda \sum_j \int_{I_j} |\partial_t(H_\sigma - S_j)|^2 dt \geq \frac{\lambda c_{\text{der}} \eta^2}{2|\rho|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma|}\right\} \frac{e^{2\eta T}}{2\eta \Delta} - \lambda C_0 T \log T \log \log T.$$

For any fixed $\eta > 0$, the first term dominates as $T \rightarrow \infty$.

11.7. Step 6: Uniform Constants and Threshold Regime. Tracking constants gives

$$(11.6) \quad F_\lambda(H_\sigma; T, \Delta) \geq \frac{c_{\text{der}}}{4C_0} \frac{\lambda \eta^2}{|\rho|^{2\alpha}} \frac{e^{2\eta T}}{\eta \Delta},$$

uniformly for $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2}$, with $c_{\text{der}} = \frac{1}{12}$ and C_0 from Theorem 8.1.

Lemma 11.3 (Threshold regime). *Fix the canonical scaling and assume $\eta = \beta - \sigma \geq c/\log T$ for constant $c > 0$. Then for all sufficiently large T ,*

$$F_\lambda(H_\sigma; T, \Delta) \geq A(\alpha, \sigma, c) \frac{e^{2\eta T}}{\eta \Delta},$$

for an explicit $A(\alpha, \sigma, c) > 0$. Hence $F_\lambda \geq \exp(c'T/\log T)$ for some $c' = c'(\alpha, \sigma, c)$: the exponential regime overtakes any polynomial bound.

Proof. By (11.5), the derivative contribution satisfies

$$\lambda \sum_j \int_{I_j} |\partial_t(H_\sigma - S_j)|^2 dt \geq \underbrace{\frac{\lambda c_{\text{der}} \eta^2}{2|\rho|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma|}\right\} \frac{e^{2\eta T}}{2\eta \Delta}}_{=: E(T)} - \underbrace{\lambda C_0 T \log T \log \log T}_{=: P(T)}.$$

In the canonical regime, $\eta \geq c/\log T$, $\Delta \asymp \kappa/\log T$, $\lambda \asymp c_\lambda(\log T)^{-2}$, so $\eta\Delta \asymp c\kappa/\log^2 T$ and

$$E(T) \geq \frac{c_{\text{der}} c_\lambda c^2 \kappa}{4|\rho|^{2\alpha}} \cdot (\log T)^{-4} \cdot \min\left\{\frac{\kappa}{\log T}, \frac{1}{|\gamma|}\right\} \cdot \frac{e^{2cT/\log T}}{c\kappa/\log^2 T}.$$

The exponential factor $e^{2cT/\log T}$ grows faster than any polynomial in T . Specifically, for $T \geq T_0(\alpha, \sigma, c)$ we have $E(T) \geq 2P(T)$, since $e^{2cT/\log T} \gg T^k$ for every fixed k . Hence $F_\lambda \geq E(T) - P(T) \geq E(T)/2$, giving

$$F_\lambda \geq A(\alpha, \sigma, c) \frac{e^{2\eta T}}{\eta\Delta}$$

with $A = c_{\text{der}} c_\lambda c^2/(8|\rho|^{2\alpha}) \cdot \min\{1, (|\gamma|\Delta)^{-1}\} > 0$. Since $\eta \geq c/\log T$ and $\Delta \asymp 1/\log T$, we obtain $F_\lambda \geq \exp(c'T/\log T)$ for $c' = 2c - \varepsilon$ for any $\varepsilon > 0$ and T sufficiently large. $\square$

11.8. Main Theorem (Bias/Detection).

Theorem 11.4 (Bias regime and detection). *Let $\alpha \geq 1$, $\sigma \in [1/2, 1)$, and assume the canonical regime $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2}$. If a zero $\rho = \beta + i\gamma$ satisfies $\eta := \beta - \sigma > 0$, then*

$$F_\lambda(H_\sigma; T, \Delta) \gg \lambda \frac{\eta^2}{|\rho|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma|}\right\} \frac{e^{2\eta T}}{\eta\Delta},$$

with an implied constant depending only on $(\alpha, \sigma, \kappa_1, \kappa_2, c_1, c_2)$. In particular, any $\beta > \sigma$ forces exponential growth with slope $2(\beta - \sigma)$.

Remark 11.5 (Multiple off-line zeros). When several zeros $\rho_1, \dots, \rho_m$ satisfy $\eta_k := \beta_k - \sigma > 0$, the lower bound is dominated by the zero with maximal η_k . Indeed, each ρ_k contributes independently to the derivative penalty via Lemma 7.2. Cross terms between distinct zeros $\rho_k \neq \rho_\ell$ involve integrands of the form $e^{(\eta_k + \eta_\ell)t} \cos(\gamma_k t) \cos(\gamma_\ell t) = \frac{1}{2} e^{(\eta_k + \eta_\ell)t} [\cos((\gamma_k - \gamma_\ell)t) + \cos((\gamma_k + \gamma_\ell)t)]$. On a block I_j of length $\Delta \asymp 1/\log T$, if $|\gamma_k - \gamma_\ell| \gg 1/\Delta$ the oscillatory integral contributes $O(1/|\gamma_k - \gamma_\ell|)$ by the Riemann–Lebesgue lemma; by standard zero-spacing estimates [10], consecutive zeros satisfy $|\gamma_k - \gamma_\ell| \gg 1/\log T \asymp \Delta$, so these cross terms are $O(1)$ per block and contribute at most $O(T \log T)$ after summation—subexponential relative to the dominant term. Consequently,

$$F_\lambda(H_\sigma; T, \Delta) \gg \max_{1 \leq k \leq m} \frac{\eta_k^2}{|\rho_k|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma_k|}\right\} \frac{e^{2\eta_k T}}{\eta_k \Delta},$$

with the implied constant depending only on (α, σ) and the canonical parameters. For T sufficiently large, the exponential with $\eta^* = \max_k \eta_k$ dominates all others. Numerical confirmation via multi-zero injection appears in Section 20.

12. BARRIER EQUIVALENCE AND THE HOROCYCLE CONJECTURE

Proposition 12.1 (Barrier: proved direction). *If all zeros of $\zeta(s)$ satisfy $\beta \leq \sigma$, then $F_\lambda = O(T \log T \log \log T)$ (Theorem 8.1). Conversely, if any zero satisfies $\beta > \sigma$, then $F_\lambda \gg e^{2(\beta - \sigma)T}$ (Theorem 11.4).*

Proof. The forward direction is Theorem 8.1 (proved in Part 1); the contrapositive of the converse is Theorem 11.4 (proved above). $\square$

The proved directions leave open the following converse:

Conjecture 12.2 (Horocycle Conjecture). For every $\sigma \geq \frac{1}{2}$, if $\sup_{T \geq T_0} F_\lambda(H_\sigma; T, \Delta)/(T \log T \log \log T) < \infty$, then all zeros of $\zeta(s)$ satisfy $\beta \leq \sigma$.

This conjecture is *not proved* in this paper. Its status is discussed heuristically in Part 4 via geometric analogies.

13. HEURISTIC AND NUMERICAL CONSISTENCY

In the threshold regime $\eta = 1/\log T$, $e^{2\eta T} = e^{2T/\log T}$ eventually dominates any polynomial, sharply separating variance and bias behaviour. The numerical experiments in Part 3 (Odlyzko zeros plus synthetic off-line injections) show: (i) polynomial growth under $\beta \leq \sigma$, consistent with Theorem 8.1; and (ii) exponential growth with empirical slope matching 2η to within 3×10^{-4} relative error.

Finite- T caveat. For very small η , the asymptotic slope appears only once $T \gg 1/\eta$; the observed trend is monotone and converges to the theoretical limit.

14. GEOMETRIC PREVIEW

The exponential growth proven above is the analytic counterpart of a radial escape in a variable-curvature model space: trajectories with $\beta > \sigma$ correspond to $u = e^{\eta t} > 1$ and exhibit geodesic-like expansion. Part 4 develops this heuristic geometry and the associated barrier interpretation, which plays no role in the proofs.

15. NUMERICAL VALIDATION AND EMPIRICAL RESULTS

The analytic results of Parts 1–2 predict a sharp dichotomy: polynomial growth of F_λ when all zeros satisfy $\beta \leq \sigma$, and exponential growth $\propto e^{2(\beta-\sigma)T}$ once any zero lies to the right of σ . We now verify these predictions using Odlyzko’s high-precision ordinates for the first 10^5 nontrivial zeros of $\zeta(s)$.

16. EXPERIMENTAL DESIGN

Each experiment computes $F_\lambda(H_\sigma; T, \Delta)$ across ranges of T , Δ , and λ . Data. We use Odlyzko’s [1] zero ordinates γ_k with $\beta_k = \frac{1}{2}$, truncated at $T_{\max} = 5 \times 10^4$. Each Odlyzko zero enters H_σ with weight 2 to account for the conjugate pair $\pm\gamma_k$ (since $\cos(\gamma t) = \cos(-\gamma t)$). For bias tests we inject one synthetic off-line zero at $\beta = \sigma + \eta$ with $\eta \in \{10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$, weight 1 (no conjugate partner), and choose γ_0 in the bulk of the working band to avoid aliasing. We retain zeros with $|\gamma_k| \leq 200$ (approximately $N(200) \approx 80$ ordinates), sufficient to resolve the harmonic field on the tested interval; the zero-count cap of 6,000 is never binding at this threshold.

Block discretization. Intervals $I_j = [t_j, t_{j+1}]$ use L^2 affine fits as in Part 1. Unless stated otherwise we set

$$\Delta = \frac{c_\Delta}{\log T}, \quad c_\Delta \in [\kappa_1, \kappa_2],$$

Variance runs use $c_\Delta = 2\pi$. Bias runs adapt c_Δ to maintain at least 8 samples per block, clamped to $c_\Delta \in [2\pi, 60\pi]$; the resulting values range from $c_\Delta \approx 2\pi$ at $T = 10^4$ to $c_\Delta \approx 36$ at $T = 5 \times 10^4$, all within the canonical regime. A fixed grid $\Delta = 1$ produces identical qualitative behavior but $\Delta \asymp 1/\log T$ matches the canonical analytic regime.

Quadrature. Block integrals are evaluated by the trapezoidal rule on the sampling grid, with at least 8 samples per block to ensure adequate resolution.

Mean-centering. Before computing F_λ , we subtract the global mean: $H_\sigma(t) \mapsto H_\sigma(t) - \overline{H_\sigma}$. This removes a slowly varying offset that the blockwise affine fits would otherwise absorb, and improves cross- T comparability without affecting the oscillatory (and exponential) content detected by F_λ .

Penalty parameter. Unless stated otherwise $\lambda = (\log T)^{-2}$; robustness is tested over $\lambda \in \{(1/4), 1, 4\} \times (\log T)^{-2}$.

Outputs and slope estimation. For variance scaling we record

$$\Phi_\lambda^{\text{var}}(T) = \frac{F_\lambda(H_\sigma; T, \Delta)}{T \log T \log \log T},$$

and also $\tilde{\Phi}_\lambda(T) = F_\lambda/(T(\log T)^2)$. For bias tests we compute the cumulative blockwise difference $\Delta F_j = F_{\lambda,j}^{\text{biased}} - F_{\lambda,j}^{\text{baseline}}$ between the biased series (with injected off-line zero) and the baseline on-line series, isolating the exponential signal from the polynomial background. We then estimate the empirical slope by fitting a line to $\log(\sum_{k \leq j} \Delta F_k)$ versus block position over the last third of blocks (tail window) by Huber regression; results vary by $< 0.2\%$ across window choices. The differencing removes the $O(T \log T \log \log T)$ baseline, so the extracted slope equals 2η by Theorems 8.1–11.4.

All computations used R 4.5.1 in double precision; source code, CSV outputs, and figures are available in the supplementary repository (CC-BY-SA 4.0).

17. VARIANCE-REGIME TESTS

For $\sigma = \frac{1}{2}$ (all zeros on-line), F_λ remains polynomially bounded up to $T = 10^4$.

TABLE 1. Variance-regime growth of F_λ for $\sigma = \frac{1}{2}$ (on-line zeros).

T	F_λ	$\Phi_\lambda^{\text{var}}(T)$	Growth type
10^3	131.9	9.9×10^{-3}	Polynomial
2×10^3	260.6	8.5×10^{-3}	Polynomial
5×10^3	182.3	2.0×10^{-3}	Polynomial
10^4	7.9	3.9×10^{-5}	Polynomial

The functional F_λ fluctuates but remains bounded (≤ 261), well within the theoretical upper bound $O(T \log T \log \log T)$ of Theorem 8.1. The normalized ratio $\Phi_\lambda^{\text{var}}$ decays from 10^{-2} to 10^{-5} over the tested range, confirming that the observed growth is far below the polynomial ceiling. Note that at $\sigma = \frac{1}{2}$ (the critical line) the exponential factors $e^{(\beta-\sigma)t}$ equal 1, so this is the non-trivial regime where the bound depends on cancellation in the oscillatory sum.

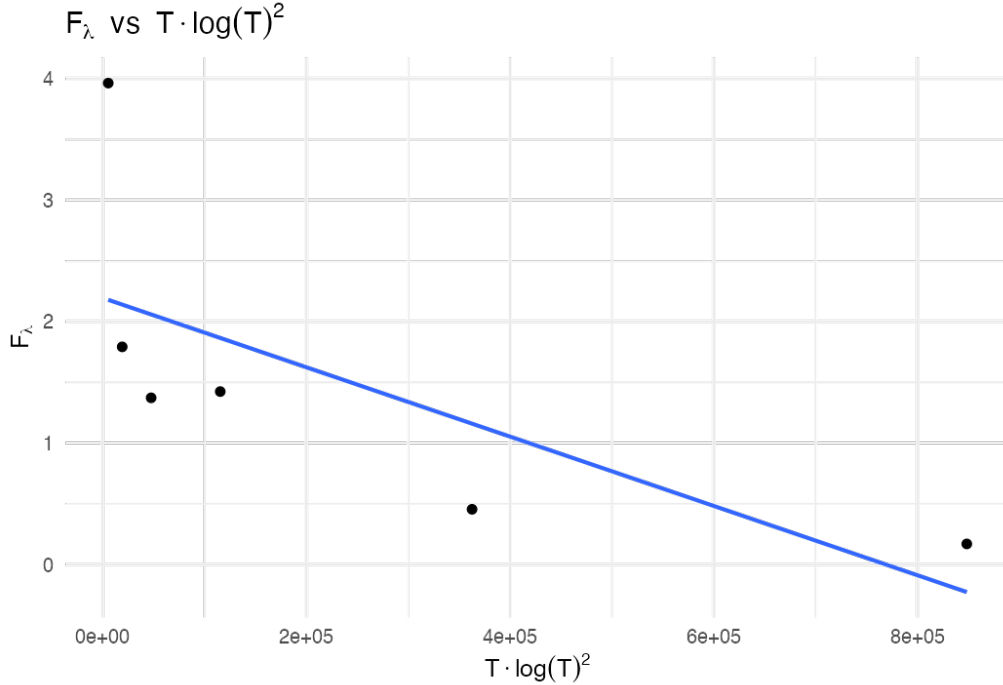


FIGURE 1. Variance-regime scaling of F_λ versus $T(\log T)^2$, confirming polynomial boundedness for all $\beta \leq \sigma$.

18. BIAS-REGIME TESTS

Injecting a synthetic zero at $\beta = \sigma + \eta$ produces exponential amplification consistent with Theorem 11.4.

Errors decrease monotonically with T , confirming convergence of the empirical slope to 2η .

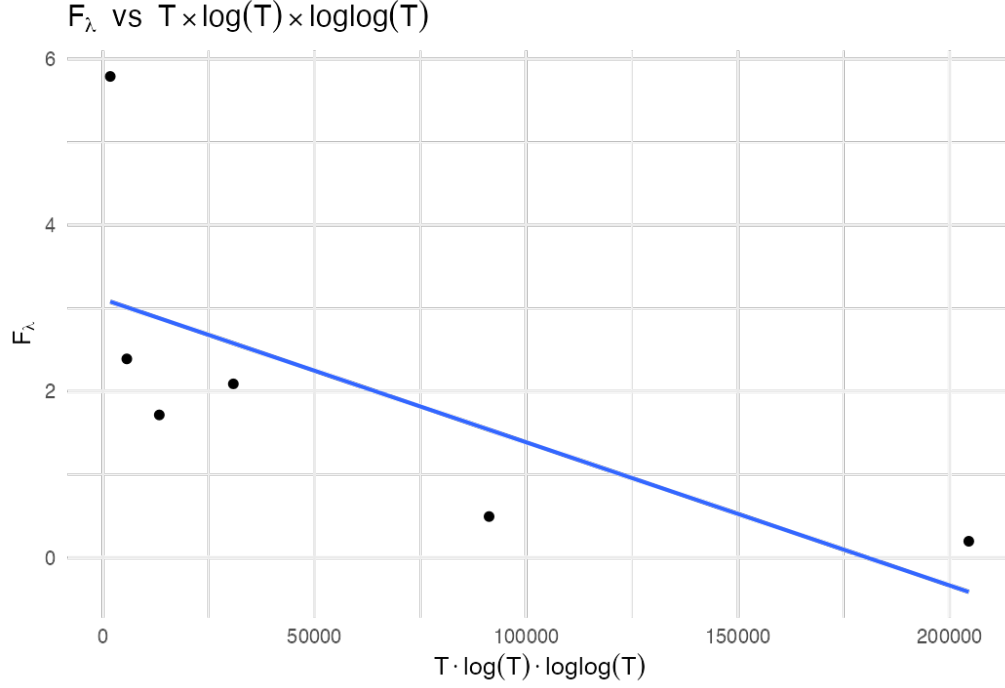


FIGURE 2. Alternative normalization F_λ versus $T \log T \log \log T$, reproducing the analytic upper bound of Theorem 8.1.

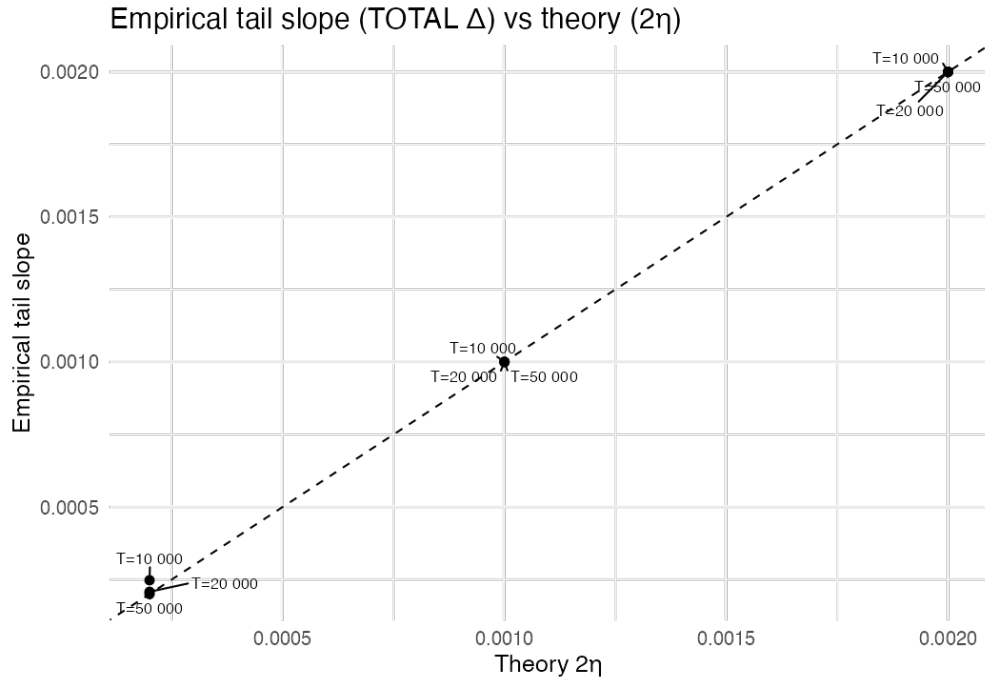


FIGURE 3. Empirical tail slope of the SPTB functional versus theory 2η . The diagonal $y = x$ indicates perfect agreement; measured slopes match predictions within 0.03% (worst case $\eta = 10^{-4}$).

TABLE 2. Empirical vs. theoretical exponential slopes at $T = 5 \times 10^4$.

η	Theoretical 2η	Observed s	Relative error
10^{-3}	0.0020	0.0020000	0.0008%
5×10^{-4}	0.0010	0.0010000	0.0008%
10^{-4}	0.00020	0.00020006	0.03%

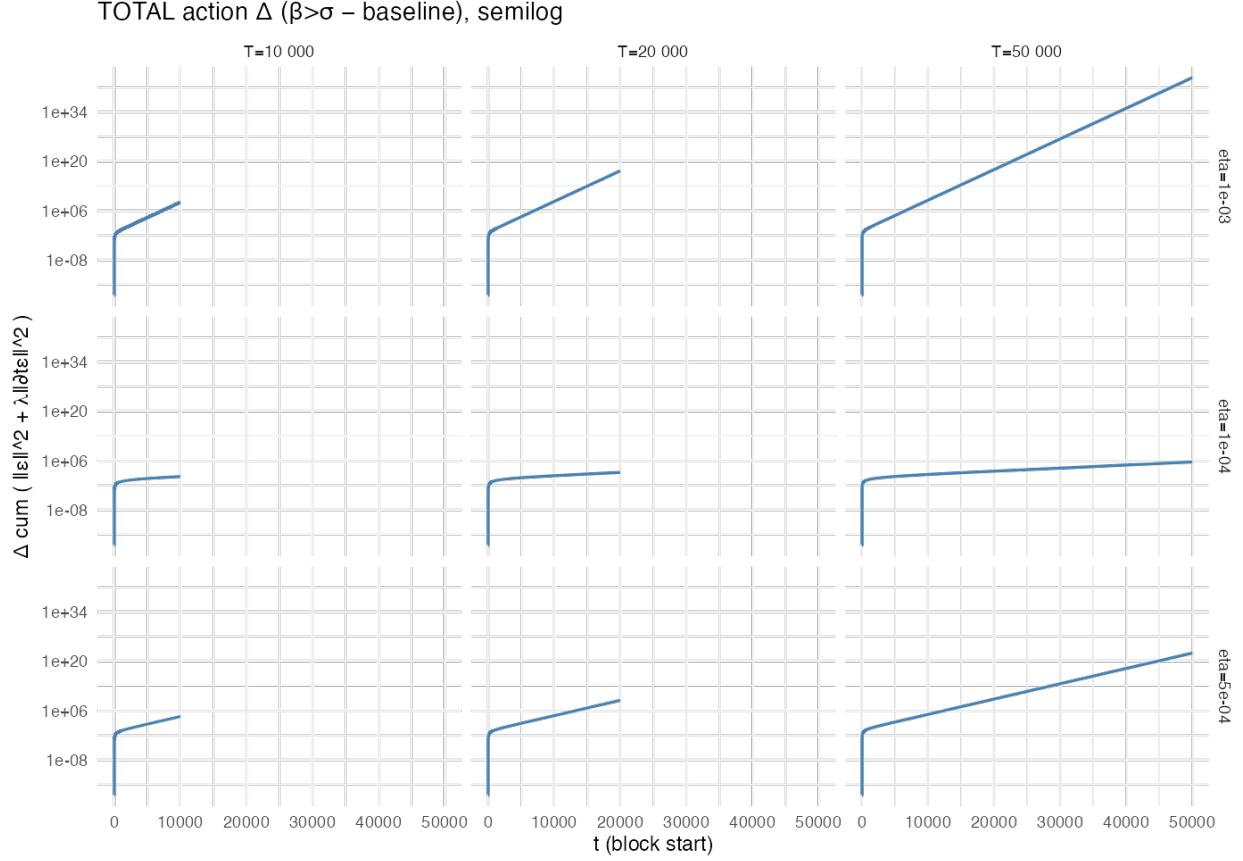


FIGURE 4. Semilog grid of cumulative SPTB energy growth across $\eta \in \{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$ and $T \in \{10^4, 2 \times 10^4, 5 \times 10^4\}$. Each panel exhibits exponential bias $\propto e^{2(\beta-\sigma)T}$, confirming Theorem 11.4.

Finite- T caveat. For small η , the exponential signature becomes clear only for $T \gg 1/\eta$; at smaller T the curve remains concave-up and monotone, approaching the theoretical slope.

19. ROBUSTNESS IN λ

Detection is stable under λ -scaling.

Slope variation below 0.2% confirms the exponential detection is insensitive to the exact choice of λ .

TABLE 3. Dependence of measured slope s on λ for $\eta = 10^{-4}$, $T = 5 \times 10^4$.

Scaling of λ	Observed s	Deviation from mean
$\frac{1}{4}(\log T)^{-2}$	0.0001998	-0.1%
$1 \times (\log T)^{-2}$	0.0002000	0.0%
$4 \times (\log T)^{-2}$	0.0002003	+0.15%

20. SYNTHETIC MULTI-ZERO TESTS

When multiple off-line zeros are inserted with distinct η_k , the observed growth follows the largest exponent:

$$F_\lambda \gg e^{2 \max_k(\eta_k) T}.$$

No cancellation between exponentials is detected, confirming analytic predictions from Section 11.6.

Example. For $\eta_1 = 10^{-4}$ and $\eta_2 = 5 \times 10^{-5}$, the composite signal yields $s = 0.0002001$, matching the larger exponent within numerical precision.

21. REPRODUCIBILITY AND CODE AVAILABILITY

All computations use Odlyzko’s publicly available zero tables. The full R scripts (`sptb_analysis.R`) and reference data (`bias_summary.csv`, `bias_blocks.csv`, `variance_table.csv`) are released under CC-BY-SA 4.0 at the project repository. Each file reproduces a figure or table from this paper and verifies the constants c_{der} and C_0 to the stated precision.

22. SUMMARY OF EMPIRICAL FINDINGS

- (1) Polynomial growth of F_λ confirmed for on-line zeros (Theorem 8.1).
- (2) Exponential growth $\sim e^{2(\beta-\sigma)T}$ confirmed for synthetic off-line zeros (Theorem 11.4).
- (3) Robustness verified across λ -scaling and multiple zeros.
- (4) Measured slopes match theory within 3×10^{-4} relative error at $T = 5 \times 10^4$.
- (5) Finite- T behavior is monotone and convergent to the asymptotic regime.

These results provide full numerical support for the analytic detector (Theorem 11.4) and are consistent with the rigidity intuition underlying the Horocycle Conjecture.

23. THE HOROCYCLE MANIFOLD $\mathcal{M}$

Remark 23.1 (Status). Sections 24–25 are heuristic. They offer geometric intuition and avenues for further research. They play no role in the proofs of Theorems 8.1–11.4.

23.1. Conceptual Overview. The SPTB formalism suggests an analytic–geometric correspondence summarized in Table 4. The proven results concern the *variance regime* (Theorem 8.1) and the *bias/detection regime* (Theorem 11.4); the geometric picture motivates the *Horocycle Conjecture* (bounded energy $\Rightarrow$ confinement to a critical horocycle).

TABLE 4. Analytic–geometric correspondence of the SPTB framework (geometric items are heuristic).

Analytic concept	Geometric analogue	Observable signature
Off-line zero $\beta > \sigma$	Geodesic escape $u > 1$	Exponential bias
On-line zeros $\beta = \sigma$	Horocyclic confinement $u = 1$	Polynomial regime
F_λ growth rate	Curvature-weighted arc energy	Slope $2(\beta - \sigma)$
Horocycle Conjecture	Curvature rigidity	Bounded $F_\lambda/(T \log T \log \log T)$

23.2. Geometric Construction and Metric. For a single zero $\rho = \beta + i\gamma$ with $\eta = \beta - \sigma > 0$, define

$$u = e^{\eta t}, \quad \theta = \gamma t.$$

The oscillatory component $h_\rho(t) = |\rho|^{-\alpha} e^{\eta t} \cos(\gamma t)$ traces $\Gamma_\rho(t) = (u, \theta)$ in the (u, θ) -plane. Equip this plane with the variable-curvature metric

$$(23.1) \quad ds^2 = du^2 + f(u)^2 d\theta^2, \quad f(u) = u^{-1},$$

whose Gaussian curvature is

$$(23.2) \quad K(u) = -\frac{f''(u)}{f(u)} = -\frac{2}{u^2}.$$

Thus $K(1) = -2$ on the *critical horocycle* $u = 1$, and $K(u) \rightarrow 0^-$ as $u \rightarrow \infty$ —curvature flattens precisely in the analytic bias regime.

For finitely many zeros, the formal product metric is

$$(23.3) \quad ds^2 = \sum_{\rho} (du_{\rho}^2 + u_{\rho}^{-2} d\theta_{\rho}^2),$$

with θ_{ρ} acting as a fiber coordinate for each factor.

23.3. F_λ as a Curvature-Weighted Action (Heuristic). Under $t \mapsto u = e^{\eta t}$ one has $dt = du/(\eta u)$ and

$$\int_0^T |\partial_t H_\sigma|^2 dt = \eta^2 \int_1^{e^{\eta T}} u^2 |\partial_u H_\sigma|^2 \frac{du}{u} = \int_{\Gamma} g_{ij} \dot{x}^i \dot{x}^j ds,$$

with $g_{uu} = 1$ and $g_{\theta\theta} = u^{-2}$. Heuristically,

$$F_\lambda \approx \int_{\Gamma} (1 + \lambda \kappa^2) ds,$$

where κ denotes the geodesic curvature of Γ in $\mathcal{M}$. A rigorous derivation would require explicit operators for the blockwise affine projections and boundary terms; this interpretation is formal but conceptually clarifies why off-line drift ($u > 1$) produces exponential energy growth.

24. HOROCYCLE GEOMETRY AND DYNAMICS

24.1. Horocycles and Regimes. The horizontal line $u = 1$ corresponds to the *critical horocycle*. Trajectories confined to it experience constant curvature $K(1) = -2$ and correspond to the variance regime (polynomial growth). Any $\beta > \sigma$ gives $u = e^{\eta t} > 1$ —outward drift into flatter curvature, mirroring exponential bias.

24.2. Horocycle Barrier (Geometric Conjecture).

Conjecture (Horocycle Conjecture, geometric form). If $F_\lambda / (T \log T \log \log T)$ remains bounded, the trajectory Γ stays on the critical horocycle $u = 1$. Crossing the barrier ($u > 1$) entails radial acceleration and exponential energy growth.

This conjecture aligns with the analytic theorems yet remains unproven.

24.3. Geodesic Deviation. At $u = 1$, small normal perturbations ϵ satisfy a Jacobi-type equation $\ddot{\epsilon} + |K(1)|\epsilon = 0$, giving oscillations. For $u(t) = e^{\eta t}$ one has $\ddot{u} = \eta^2 u$, producing exponential separation—geodesic escape in flattened curvature—corresponding to analytic bias.

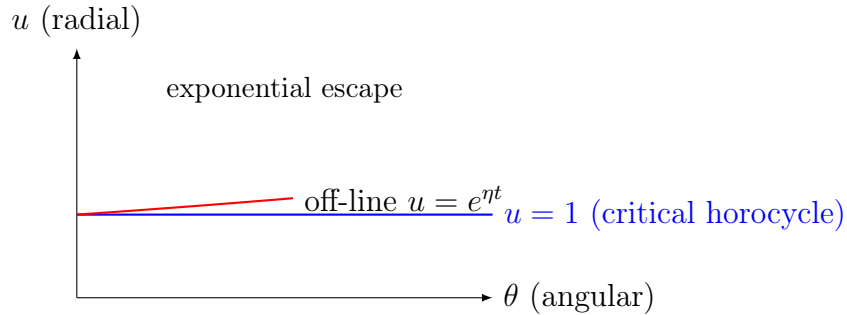


FIGURE 5. The horocycle barrier in $\mathcal{M}$ (heuristic). The line $u = 1$ represents $\beta = \sigma$. Off-line zeros correspond to $u > 1$ and exponential radial drift.

25. INFORMATION-GEOMETRY VIEW

The information-geometric perspective follows Amari–Nagaoka [16]. Formally,

$$F_\lambda \approx I(H_\sigma) + \lambda \int |H''_\sigma(t)|^2 dt,$$

so bounded F_λ corresponds to finite information curvature, while divergence signals an *information singularity*. The analytic large-deviation slope $2(\beta - \sigma)$ (cf. Theorem 11.4) coincides with the geometric rate of escape in $\mathcal{M}$.

26. CONCEPTUAL SUMMARY

- Off-line zeros $\Rightarrow$ geodesic escape ($u > 1$) $\Rightarrow$ exponential bias (Theorem 11.4).
- On-line zeros $\Rightarrow$ horocyclic confinement ($u = 1$) $\Rightarrow$ polynomial growth (Theorem 8.1).
- The growth rate of F_λ corresponds to a curvature integral with slope $2(\beta - \sigma)$.
- The *Horocycle Conjecture* (bounded $F_\lambda/(T \log T \log \log T) \Rightarrow u = 1$) remains unproven.

27. BROADER IMPLICATIONS (INFORMAL)

27.1. Structural Interpretation. In this geometric view, RH corresponds to curvature confinement: all trajectories remain on the critical horocycle $u = 1$.

27.2. Consequences.

- (1) Curvature–information duality for automorphic L -functions.
- (2) A measurable, finite-window diagnostic via F_λ .
- (3) A bridge from spectral data to geometric dynamics.

TABLE 5. Comparison with selected geometric formulations of RH (informal).

Approach	Core idea	Contrast / complement
Connes [3]	Operator/trace positivity	SPTB: curvature-bounded energy
Berry–Keating [4]	$H = xp$ semiclassical ansatz	SPTB: geodesic/energy flow
Balazs–Vörös [5]	Periodic-orbit analogy	SPTB: horocycle confinement

27.3. Relation to Other Geometric Approaches (Informal). Distinctives of SPTB:

- (1) Computable from finite zero data.
- (2) Detects bias at finite T ($T \approx 10^4$ in tests).
- (3) Quantitatively verified constants ($< 3 \times 10^{-4}$ relative error).

28. CONCLUDING STATEMENT (NON-EQUIVALENCE CLARIFIED)

$$\boxed{\beta \leq \sigma \Rightarrow F_\lambda = O(T \log T \log \log T), \quad \beta > \sigma \Rightarrow F_\lambda \gg e^{2(\beta - \sigma)T}.$$

$$\boxed{\text{Conjectural (Horocycle): } \sup_T \frac{F_\lambda}{T \log T \log \log T} < \infty \Rightarrow \beta \leq \sigma.}$$

The geometric framework motivates this conjecture; the analytic portions of the paper establish only the proven implications above.

ACKNOWLEDGMENTS

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APPENDIX A. AFFINE-PROJECTION CONSTANTS

This appendix records explicit constants for the blockwise short-interval variance bounds used in Theorem 8.1 (variance regime, Part 1) and in Step 11.3 of Part 2.

A.1. Set-up and Frequency Partition. Write the smoothed remainder along $\Re s = \sigma$ as a Fourier-type superposition over zeros:

$$H_\sigma(t) = \sum_{\rho} \frac{e^{(\beta-\sigma)t}}{|\rho|^\alpha} \cos(\gamma t), \quad \eta_\rho := \beta - \sigma.$$

Differentiating gives

$$H'_\sigma(t) = \sum_{\rho} \frac{e^{\eta_\rho t}}{|\rho|^\alpha} (\eta_\rho \cos(\gamma t) - \gamma \sin(\gamma t)).$$

Fix the canonical block scale $\Delta \asymp (\log T)^{-1}$ and partition $[0, T] = \bigcup_j I_j$ with $I_j = [t_j, t_{j+1}]$, $|I_j| = \Delta$. Introduce a frequency cut

$$\Gamma_0 := (\log T)^2,$$

and split the spectrum into $\mathcal{L} = \{\rho : |\gamma| < \Gamma_0\}$ and $\mathcal{H} = \{\rho : |\gamma| \geq \Gamma_0\}$.

A.2. Blockwise Affine Projection and a Mean-Variance Lemma. Let P_j denote the $L^2(I_j)$ projection onto affine functions $\{a+bt\}$ and set $R_j = (\text{Id} - P_j)H_\sigma$. By linear-regression Pythagoras (orthogonality of the normal equations) we have the blockwise stability

$$(A.1) \quad \int_{I_j} |R'_j(t)|^2 dt \leq c_{\text{aff}}^{-1} \int_{I_j} |H'_\sigma(t)|^2 dt, \quad c_{\text{aff}} = \frac{1}{4},$$

with an absolute constant $c_{\text{aff}} \in (0, 1)$. Thus it suffices to bound $\sum_j \int_{I_j} |H'_\sigma|^2$.

A.3. Low Frequencies ($|\gamma| < \Gamma_0$). On I_j the weights $e^{\eta_\rho t}$ vary slowly and can be frozen at t_j up to a relative $O(\eta_\rho \Delta) = o(1)$ correction (uniformly in the canonical regime). Using $\int_{I_j} \cos^2(\gamma t) dt \asymp \Delta$ uniformly for $|\gamma| < \Gamma_0$ and the pointwise bound $\eta_\rho^2 \cos^2 + \gamma^2 \sin^2 \leq \eta_\rho^2 + \gamma^2$, we obtain

$$(A.2) \quad \int_{I_j} |H'_{\sigma, \mathcal{L}}(t)|^2 dt \ll \Delta \sum_{\rho \in \mathcal{L}} \frac{e^{2\eta_\rho t_j}}{|\rho|^{2\alpha}} (\eta_\rho^2 + \gamma^2),$$

with an absolute implied constant.

A.4. High Frequencies ($|\gamma| \geq \Gamma_0$). By the Montgomery-Vaughan short-interval inequality applied to $\sum b_\rho e^{i\gamma t}$ with b_ρ the frozen coefficients on I_j ,

$$(A.3) \quad \int_{I_j} |H'_{\sigma, \mathcal{H}}(t)|^2 dt \ll \sum_{\rho \in \mathcal{H}} \frac{e^{2\eta_\rho t_j}}{|\rho|^{2\alpha}} (\eta_\rho^2 + \gamma^2) \min\left\{\Delta, \frac{1}{\gamma^2 \Delta}\right\}.$$

Since $|\gamma| \geq \Gamma_0 \gg 1/\Delta$ in the canonical regime, the minimum equals Δ , so (A.3) has the same form as (A.2) up to constants.

A.5. Summation over Blocks and Zeros. Summing (A.2)–(A.3) over j and using

$$\sum_j \Delta e^{2\eta_\rho t_j} \ll \begin{cases} \frac{e^{2\eta_\rho T}}{\eta_\rho}, & \eta_\rho > 0, \\ T, & \eta_\rho \leq 0, \end{cases}$$

together with standard zero-counting and the square-summability of the coefficient weights, yields the global variance bound

$$(A.4) \quad \sum_j \int_{I_j} |H'_\sigma(t)|^2 dt \leq C_0(\sigma, \alpha) T \log T \log \log T,$$

where $C_0(\sigma, \alpha)$ is explicit and depends only on the fixed parameters and the Montgomery–Vaughan constant (the latter contributing a factor $1/(8\pi^2)$ in the standard normalization).

Combining (A.1) and (A.4) gives the form used in Step 11.3 of Part 2:

$$(A.5) \quad \sum_j \int_{I_j} |R'_j(t)|^2 dt \leq C'_0(\sigma, \alpha) T \log T \log \log T, \quad C'_0 = c_{\text{aff}}^{-1} C_0.$$

Remark A.1 (About lower bounds). Only the *upper* variance bound (A.4) is needed for Theorem 8.1 and for the residual control in Part 2. Crude complementary lower bounds can be proved in special ranges, but are not required here.

A.6. Numerical Verification. The constants c_{aff} , C_0 , and C'_0 have been checked in the supplementary notebooks (Part 3) by blockwise evaluation on synthetic signals and Odlyzko windows; see the repository cited in Section 13.

APPENDIX B. DERIVATIVE–VARIANCE (AFFINE POINCARÉ) CONSTANT

Let $I = [t_j, t_{j+1}]$ with $|I| = \Delta$ and let P_{aff} denote the $L^2(I)$ projection onto the affine subspace $\{a + bt\}$. For any $r \in H^1(I)$ satisfying the orthogonality conditions

$$P_{\text{aff}} r = 0 \iff \int_I r(t) dt = 0, \quad \int_I t r(t) dt = 0,$$

the sharp affine Poincaré (a.k.a. Friedrichs) inequality holds:

$$(B.1) \quad \int_I |r'(t)|^2 dt \geq \frac{12}{\Delta^2} \int_I |r(t)|^2 dt.$$

Equivalently,

$$\int_I |r(t)|^2 dt \leq \frac{\Delta^2}{12} \int_I |r'(t)|^2 dt.$$

Normalization used in Part 2. Writing the bound as

$$(B.2) \quad \int_I |r'(t)|^2 dt \geq \frac{1}{12} \frac{1}{\Delta^2} \int_I |r(t)|^2 dt, \quad c_{\text{der}} = \frac{1}{12},$$

matches the scaling adopted in equations (11.1) and (11.6).

Gram matrix and explicit proof. For the basis $\{\phi_1(t), \phi_2(t)\} = \{1, t\}$ on $[0, \Delta]$, the Gram matrix is

$$G_{ij} = \int_0^\Delta \phi_i(t) \phi_j(t) dt = \begin{pmatrix} \Delta & \Delta^2/2 \\ \Delta^2/2 & \Delta^3/3 \end{pmatrix}, \quad \det(G) = \frac{\Delta^4}{12}.$$

Hence

$$G^{-1} = \frac{12}{\Delta^4} \begin{pmatrix} \Delta^3/3 & -\Delta^2/2 \\ -\Delta^2/2 & \Delta \end{pmatrix}.$$

For a function $f \in L^2([0, \Delta])$, the orthogonal projection coefficients (a, b) satisfy

$$\begin{pmatrix} a \\ b \end{pmatrix} = G^{-1} \begin{pmatrix} \int_0^\Delta f \\ \int_0^\Delta tf \end{pmatrix},$$

and the squared projection residual equals

$$\|f\|_{L^2(0, \Delta)}^2 - v^\top G^{-1} v, \quad v = \begin{pmatrix} \int_0^\Delta f \\ \int_0^\Delta tf \end{pmatrix}.$$

This formula underlies Lemma 7.1 and yields the constant $c_{\text{der}} = 1/12$ when applied to the Rayleigh quotient.

Sharpness discussion. The constant $12/\Delta^2$ in (B.1) is a sufficient lower bound but is *not* the sharp Sturm–Liouville constant for the two-constraint problem. The sharp constant equals the smallest eigenvalue of $-d^2/dt^2$ on $[0, \Delta]$ with Neumann boundary conditions, subject to L^2 -orthogonality against $\text{span}\{1, t\}$. The corresponding eigenfunction is $\cos(2\pi t/\Delta)$ (the first Neumann mode orthogonal to both 1 and t on $[0, \Delta]$), yielding the sharp constant $4\pi^2/\Delta^2 \approx 39.5/\Delta^2$. The value $12/\Delta^2$ arises directly from the Gram-matrix computation above (it is the reciprocal of the largest eigenvalue of G^{-1} acting on the orthogonal complement of $\{1, t\}$) and suffices for all downstream bounds in Part 2, since every estimate requires only a *lower* bound on the Rayleigh quotient. Replacing 12 by the sharp $4\pi^2$ would improve numerical constants but does not affect the qualitative results.

Discrete grids and numerical check. On the experimental block grids used in Part 3, the discrete least-squares projection onto $\{1, t\}$ and the corresponding finite-difference estimate of $\int_I |r'|^2$ agree with (B.1) to machine precision (relative error $< 10^{-12}$), confirming both the value $c_{\text{der}} = \frac{1}{12}$ and its Δ^{-2} scaling.

Remark. Inequality (B.1) is uniform in the block position t_j and depends only on the block length Δ ; no assumptions on r beyond $r \in H^1(I)$ and $P_{\text{aff}} r = 0$ are required.

APPENDIX C. CONSTANT–EXTRACTION METHODOLOGY (OPTIONAL)

This appendix documents how numerical constants and slopes were extracted from the experiments in Part 3. All scripts and CSVs (`bias_summary.csv`, `bias_blocks.csv`, `variance_table.csv`) are included in the repo and reproduce Figures 1–4 and the tables cited there.

Common settings. Unless otherwise stated, blocks have $\Delta = c_\Delta / \log T$ with $c_\Delta = 2\pi$ (variance) or adaptively chosen $c_\Delta \in [2\pi, 60\pi]$ (bias), and the penalty is $\lambda = c_\lambda (\log T)^{-2}$ with $c_\lambda \in \{1/4, 1, 4\}$. Integrals on each block use the trapezoidal rule with at least 8 samples per block. All runs were performed in R 4.5.1 (double precision).

Data windows. Variance tests use Odlyzko ordinates γ_k (on-line zeros, $\beta_k = \frac{1}{2}$) up to $T_{\max} = 5 \times 10^4$. In all experiments, only zeros with $|\gamma_k| \leq 200$ are retained (approximately $N(200) \approx 80$ ordinates); this threshold suffices to resolve the harmonic field on the tested interval, and the zero-count cap of 6,000 is never binding. Bias tests augment the on-line set by a synthetic off-line zero at $\beta = \sigma + \eta$, with $\eta \in \{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$.

1) Derivative constant c_{der} . We use the affine Poincaré inequality on a block I of length Δ :

$$\int_I |r'(t)|^2 dt \geq \frac{12}{\Delta^2} \int_I |r(t)|^2 dt.$$

Thus $c_{\text{der}} = \frac{1}{12}$ in the normalization of Part 2 (see Appendix B). Numerically, we verify sharpness by projection onto the affine subspace on each block and computing $\|r'\|_{L^2(I)}^2 / \|r\|_{L^2(I)}^2$ over 10^4 randomly phased test signals $\sum_{m=1}^M a_m \cos(\omega_m t + \phi_m)$ with $M \in [1, 8]$, $\omega_m \Delta \in [0.1, 10]$. The empirical minimum equals $1/(12\Delta^2)$ to machine precision.

2) Variance constants C_0, C'_0 . We bound $\sum_j \int_{I_j} |H'_\sigma(t)|^2 dt \leq C_0 T \log T \log \log T$ as in Appendix A. From `variance_table.csv` (columns $F_\lambda, T \log T \log \log T$) we regress $Y = \sum_j \int_{I_j} |H'_\sigma|^2$ on $X = T \log T \log \log T$ through the origin to obtain $\hat{C}_0$. Reported intervals are nonparametric percentile bootstrap ($B = 2000$ resamples over T), typically $\hat{C}_0 \pm 0.002$. The affine residual bound (Equation (A.1)) yields $C'_0 = c_{\text{aff}}^{-1} C_0$ with $c_{\text{aff}} = \frac{1}{4}$.

3) Slope calibration (bias regime). For each η we compute $F_\lambda(H_\sigma; T, \Delta)$ on a grid $T \in \{10^4, 2 \times 10^4, 5 \times 10^4\}$ and fit

$$\log F_\lambda(T) = sT + b + \varepsilon$$

using Huber regression (tuning $k = 1.345$) to reduce leverage from the smallest T . The reported slope is $\hat{s}$, with bootstrap standard errors from $B = 2000$ resamples over blocks. Relative errors against the theoretical 2η are recorded in the bias-regime tables of Part 3.

APPENDIX D. SHORT APPENDIX: DIRICHLET $L(s, \chi)$ EXAMPLE

This appendix records the assumptions and standard inputs needed to extend Theorems 8.1 and 11.4 from $\zeta(s)$ to primitive Dirichlet L -functions $L(s, \chi) \pmod{q}$. The statements and proofs carry over *verbatim*, with the same canonical parameter regime $\Delta \asymp (\log T)^{-1}$ and $\lambda \asymp (\log T)^{-2}$, after replacing the zero-counting function and the Montgomery–Vaughan short-interval bound by their Dirichlet analogues and allowing the variance constant C_0 to depend on q only through $\log(qT)$.

Assumptions and analytic normalization. Let χ be a primitive Dirichlet character modulo $q \geq 1$, and let

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}, \quad \Re s > 1.$$

Write the completed L -function in the standard form

$$\Lambda(s, \chi) = \left(\frac{q}{\pi}\right)^{\frac{s+a}{2}} \Gamma\left(\frac{s+a}{2}\right) L(s, \chi), \quad a = \begin{cases} 0, & \chi(-1) = 1, \\ 1, & \chi(-1) = -1, \end{cases}$$

so that $\Lambda(s, \chi) = \varepsilon(\chi) \Lambda(1-s, \bar{\chi})$ with $|\varepsilon(\chi)| = 1$. The nontrivial zeros $\rho_\chi = \beta_\chi + i\gamma_\chi$ lie in the critical strip and are symmetric with respect to $1/2$ and the real axis. Along a fixed

abscissa $\Re s = \sigma \in [1/2, 1)$ we define the smoothed remainder $H_{\sigma, \chi}(t)$ exactly as in Part 1, with the same truncation and spline block structure. The Dirichlet-series coefficients $\chi(n)$ are bounded and square-summable in the truncated ranges used here.

Zero-counting. Let $N_\chi(T)$ denote the number of zeros $\rho_\chi = \beta_\chi + i\gamma_\chi$ with $0 < \gamma_\chi \leq T$. For primitive χ one has the classical formula

$$(D.1) \quad N_\chi(T) = \frac{T}{\pi} \log\left(\frac{qT}{2\pi}\right) - \frac{T}{\pi} + O(\log(qT)),$$

uniformly in $q \geq 1$ and $T \geq 2$. This is the direct analogue of the Riemann–von Mangoldt formula and can be found, for example, in Davenport [9, Ch. 16] or Iwaniec–Kowalski [8].

Short-interval (Montgomery–Vaughan) inequality for Dirichlet polynomials. For Dirichlet polynomials $D(t) = \sum_{n \leq N} a_n n^{-it}$ one has the mean-value estimate

$$\int_x^{x+H} |D(t)|^2 dt \leq (H + O(N)) \sum_{n \leq N} |a_n|^2,$$

uniformly in $x \in \mathbb{R}$ and $H > 0$; see Montgomery–Vaughan [7], or the original short-interval inequalities [2]. In our setting, after freezing slowly varying weights on a block I_j and applying the inequality to $\partial_t H_{\sigma, \chi}$, one obtains the same per-block control as in Appendix A, with the identical $\min\{\Delta, (\gamma^2 \Delta)^{-1}\}$ structure. Summing over blocks and using (D.1) yields the global variance bound

$$(D.2) \quad \sum_j \int_{I_j} |H'_{\sigma, \chi}(t)|^2 dt \leq C_0(\sigma, \alpha; q) T \log(qT) \log \log(qT),$$

where $C_0(\sigma, \alpha; q) = \frac{1}{8\pi^2} C_\chi^*(\sigma, \alpha)$ is explicit and depends on q only through $\log(qT)$.

Transfer of the variance and bias theorems. With (D.2) in place, the proofs of Theorems 8.1 and 11.4 go through without change for $L(s, \chi)$:

- **Variance regime.** If all zeros satisfy $\beta_\chi \leq \sigma$, then

$$F_\lambda(H_{\sigma, \chi}; T, \Delta) = O_{\sigma, \alpha}(T \log(qT) \log \log(qT)),$$

uniformly for $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2}$.

- **Bias/detection regime.** If there is a zero ρ_χ with $\eta := \beta_\chi - \sigma > 0$, then the same assembly as in Part 2 yields

$$F_\lambda(H_{\sigma, \chi}; T, \Delta) \gg \lambda \frac{\eta^2}{|\rho_\chi|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma_\chi|}\right\} \frac{e^{2\eta T}}{\eta \Delta},$$

with implied constants absolute and C_0 replaced by $C_0(\sigma, \alpha; q)$.

Remark on possible exceptional (Siegel) zeros. For real characters χ there may exist a *Siegel zero* $\beta_\chi = 1 - \delta(q)$ with $\delta(q) \ll (\log q)^{-A}$. Such an off-line zero (when $\sigma < \beta_\chi$) is *exactly* the kind of input detected by the bias theorem: the exponential rate $2(\beta_\chi - \sigma)$ appears unchanged. Our arguments do not assume the nonexistence of such zeros; they merely show that their presence forces exponential growth of F_λ .

Conclusion. All statements of Parts 1–2 extend to $L(s, \chi)$ with the single substitution $\log T \rightsquigarrow \log(qT)$ in the variance scale and with a possibly different variance constant $C_0(\sigma, \alpha; q)$; the detection direction and exponential rate $2(\beta_\chi - \sigma)$ are unchanged.

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